1 Explanation certificate: heart-s1-c1

This section is the explanation certificate of the heart-s1-c1 network. It is rendered from the canonical JSON record `heart-s1-c1.json` (schema 1.4.0, SHA-256 `555774df08fc19cc...`), which also holds every evidence payload, the full graphs with weights, and the explanation record. Table 1 summarises the network and its coverage.

Dataset	Heart Disease (binary, prepared)
Experiment	10seed-heart-s1, seed 1
Genome	`a2df22ce-983d-47b6-bbd6-3176d38ed5ff` (NEAT key 9674, generation 39)
Explanation	`492866e6-a73b-4465-92a8-ecad82a2adcc` (35 operations)
Train AUC / accuracy	0.9392 / 0.8802 (n = 242)
Test AUC / accuracy	0.861 / 0.7705 (n = 61)
Output activation	sigmoid
Phenotype	34 nodes (26 inputs, 7 hidden, 1 output), 30 enabled edges
Explained model state	38 nodes, 30 enabled edges
Refactoring operations	split_node 3
Function preservation	max $\lvert\Delta\,\text{output}\rvert = 3.33 \times 10^{-16}$ on the training split
Annotations	6 (5 leaf, 1 composition)
Structural coverage	1 (37 of 37 candidate nodes; 9 Null)
Compositional coverage	1
Uncovered nodes / edges	0 / 0
Missing compositions	0
Full coverage	yes
Drafted by	AI-assisted draft
Verified by	unverified
Record hash (SHA-256)	`555774df08fc19cc...`

Table 1: Explanation certificate summary: heart-s1-c1.

Drafter's summary. Essentially logistic regression (all hidden units linearised: train AUC 0.9355 vs 0.9392; L2 logistic regression 0.9355). 13 direct edges carry logit SD 4.25 of 5.63; ca enters by three parallel routes (effective +2.60 logit per SD), cp:3 only via always-on relay 1173. The one consequential nonlinearity is gate 1801 = ReLU(1.54 restecg:0 + 0.82 oldpeak - 1.52): ST depression adds risk only with a normal resting ECG (+0.0034 AUC over linear). Typical angina (cp:1) lowers risk via switch 3210, less with more vessels; a small hinge lets higher max heart rate lower risk. Same design as the 10-seed heart-s1 record (gen-31 genome), with an added typical-angina switch and heart-rate hinge.

What resisted explanation. Nothing resisted coverage; only input fan-out needed splitting (ca x3, slope:2, thalach). Four of seven hidden ReLUs (664, 1173, 2245, 2650) never clip on train, so their annotations describe affine relays (reweighting of ca, slope:2, cp:3 and the heart-rate hinge) rather than distinct concepts. 3210 is a pure cp:1 switch (on for all 22 typical-angina patients, off for the rest), so its best linear fit is almost exact. Nine unconnected inputs (sex:1, fbs:0, restecg:1, restecg:2, exang:0, exang:1, slope:1, thal:4, thal:7) listed as zero-effect root entries.

1.1 Plain-language summary

AI-assisted draft for review (unverified). Each sentence cites the claims (C01, C02, ...) that back it; every cited claim was computed on the model and the training split and passed its automatic checks. Magnitudes

are changes in predicted risk for a typical patient; confidence words follow the test-split check and n. The text describes what the model does, not the world.

Summary (drafted). For a typical patient (a 56-year-old man with no chest pain symptoms and median test results) the model predicts a risk of heart disease of about 18%; it adds up most of its inputs independently, and five parts add to that sum: two add more for each extra vessel coloured, one lowers the risk for typical angina, one lowers it for non-anginal chest pain, one acts only on the 42 of 242 training patients with a normal resting ECG and marked ST depression, and one lowers it for a higher maximum heart rate, with nothing until about 145 beats per minute, then steadily. [C29, C30, C40, C31, C43, C38, C46] These add up independently, and these lower the risk: being a woman takes a typical patient's predicted risk from about 18% to below 1% and fasting blood sugar above 120 mg/dL to about 8%; through this sum alone, compared with no chest pain symptoms, typical angina or non-anginal chest pain lowers it to about 2% and atypical angina to about 5% (chest pain type also acts through the parts below). [C10, C19, C12, C17] A reversible defect (or the undefined code 4) on the thallium scan raises it to about 55%; higher cholesterol (from about 8% at 193 mg/dL to about 45% at 308 mg/dL), higher resting blood pressure (from about 9% at 110 mm Hg to about 35% at 155 mm Hg) and older age (from about 10% at 42 years to about 25% at 66 years) raise it steadily. [C05, C07, C08, C13] Through this sum alone, each extra vessel coloured by fluoroscopy adds about the same (from about 18% at 0 to about 80% at 2 vessels) and a flat ST segment raises it to about 87%, while the maximum heart rate barely changes it; all three also act through the parts below. [C04, C03, C21] Rarer results also count (the model has little data on this, n = 15 and 17): a fixed defect on the thallium scan lowers the risk to about 3%, and through this sum alone a downsloping ST segment raises it to about 32%; the model ignores exercise-induced angina. [C14, C15, C28] This part combines chest pain type with the number of vessels coloured: through it, typical angina lowers a typical patient's predicted risk from about 18% to about 2%, and each extra vessel adds about the same, from about 18% at 0 to about 61% at 2 vessels. [C31, C30] Across the whole model, more vessels raise the predicted risk, and even more so with typical angina, each extra vessel adding about the same: going from 0 to 2 vessels moves a typical patient from below 1% to about 55% with typical angina, and from about 18% to about 99% with any other chest pain type. [C32] Across the whole model, typical angina takes a typical patient's predicted risk to below 1%. [C09] This part combines the resting ECG with ST depression on exercise and acts on 42 of 242 training patients, nearly all with a normal resting ECG, 26 of whom had heart disease; for the typical patient it acts on, the predicted risk is about 98% without it and about 99% with it. [C38] Through it, more ST depression raises the predicted risk only once it rises above about 1.0 mm, for patients with a normal resting ECG, in an uneven way: for a typical such patient from about 13% at 0 mm to about 63% at 3.5 mm. [C37] More ST depression therefore raises the predicted risk, but only if the resting ECG is normal: going from 0 to 2.8 mm moves a typical patient from about 18% to about 55% with a normal resting ECG, in an uneven way, and leaves it at about 18% otherwise. [C36] This part combines the number of vessels coloured with a flat ST segment: through it, each extra vessel adds about the same, from about 18% at 0 to about 36% at 2 vessels for a typical patient, and a flat ST segment raises the risk from about 18% to about 23%. [C40, C41] Across the whole model, each extra vessel takes a typical patient from about 18% at 0 to about 99% at 2 vessels, about the same for each, flattening out near the top because the predicted risk cannot go much higher, and a flat ST segment takes it to about 91%. [C01, C02] This part reads only non-anginal chest pain: through it, non-anginal chest pain lowers a typical patient's predicted risk from about 18% to about 8%. [C43] Across the whole model, non-anginal chest pain takes it to about 1%. [C11] This part reads only the maximum heart rate: a higher maximum heart rate lowers the predicted risk with nothing until about 145 beats per minute, then steadily, for a typical patient from about 19% at 118 to about 14% at 175 beats per minute. [C45, C46] Across the whole model the same holds, from about 19% at 118 to about 14% at 175 beats per minute. [C20]

The whole model. The model adds up its inputs into one score: the vessels coloured (each extra vessel adding about the same), the ST segment, chest pain type and sex each move the predicted risk on their own, five parts add to that sum, and exercise-induced angina is ignored. [C29, C01, C02, C10, C28] *Evidence.* It is made up of the inputs that feed the score directly and five parts: vessels with a flat ST segment, the non-anginal chest pain relay, ST depression with a normal ECG, the maximum heart rate part and typical angina with vessels (each explained below); the score is their sum, which was checked on all 242 training patients. [C29] Changing one input at a time across the whole model, each extra vessel adds about the same, taking a typical patient's predicted risk from about 18% at 0 to about 99% at 2 vessels, flattening out near the top because the predicted risk cannot go much higher, against about 80% through the sum alone. [C01, C04] More ST depression raises it in an uneven way across the whole model, from about 18% at 0 mm to about 55% at 2.8 mm; it enters the score only through the part with a normal ECG. [C06] The model ignores exercise-induced angina. [C28]

Typical angina with vessels. This part lowers the predicted risk for typical angina and adds risk for each extra vessel coloured, about the same for each, with the vessels counting more for patients with typical angina. [C31, C30, C32] *Evidence.* Through it, typical angina takes a typical patient's predicted risk from about 18% to about 2%, and each extra vessel adds about the same, from about 18% at 0 to about 61% at 2 vessels. [C31, C30] Across the whole model, going from 0 to 2 vessels moves a typical patient from below 1% to about 55% with typical angina, each extra vessel adding about the same, and from about 18% to about 99% with any other chest pain type. [C32] Its formula reproduces the part exactly on all 242 training patients. [C33]

ST depression with a normal ECG. This part raises the predicted risk with ST depression on exercise, in an uneven way, only for patients with a normal resting ECG, once it rises above about 1.0 mm. [C37, C36] *Evidence.* It acts on 42 of 242 training patients, nearly all with a normal resting ECG, 26 of whom had heart disease; for the typical patient it acts on, the predicted risk is about 98% without it and about 99% with it. [C38] Through it, more ST depression raises the predicted risk only once it rises above about 1.0 mm, in an uneven way: for a typical patient with a normal resting ECG from about 13% at 0 mm to about 63% at 3.5 mm. [C37] Going from 0 to 2.8 mm of ST depression moves a typical patient from about 18% to about 55% with a normal resting ECG, in an uneven way, and leaves it at about 18% otherwise. [C36] Its formula reproduces the part exactly on all 242 training patients. [C39]

Vessels with flat ST segment. This part adds risk for each extra vessel coloured, about the same for each, and for a flat ST segment. [C40, C41] *Evidence.* Through it, each extra vessel adds about the same, from about 18% at 0 to about 36% at 2 vessels for a typical patient, and a flat ST segment raises the risk from about 18% to about 23%. [C40, C41] Its formula reproduces the part exactly on all 242 training patients. [C42]

Non-anginal chest pain relay. This part lowers the predicted risk for patients with non-anginal chest pain. [C43] *Evidence.* Through it, non-anginal chest pain takes a typical patient's predicted risk from about 18% to about 8%. [C43] Its formula reproduces the part exactly on all 242 training patients. [C44]

Maximum heart rate. This part lowers the predicted risk for a higher maximum heart rate: nothing until about 145 beats per minute, then steadily. [C45, C46] *Evidence.* Through it, a higher maximum heart rate takes a typical patient from about 19% at 71 to about 13% at 182 beats per minute, with nothing until about 145 beats per minute, then steadily. [C46] Its formula reproduces the part exactly on all 242 training patients. [C47]

Worth checking against what is known. Worth checking: patients with typical angina or non-anginal chest pain get the lowest predicted risk of all chest pain types (below 1% and about 1% for a typical patient), below patients with no chest pain symptoms. [C09, C11] Worth checking: the model treats a fixed defect on the thallium scan as lowering the risk compared with a normal scan, from only 15 training patients. [C14] Worth checking: the model treats fasting blood sugar above 120 mg/dL as lowering the risk of heart disease (for a typical patient from about 18% to about 8%). [C19] Worth checking: going from 0 to 2 vessels coloured moves a typical patient's predicted risk from about 18% to about 99%. [C01] Worth checking: ST depression on exercise raises the predicted risk only when the resting ECG is normal. [C36] Worth checking: the model ignores exercise-induced angina. [C28]

Claims.

Claim	What is checked	Risk, typical patient	Estimate (log-odds; OR); range across patients	n	Test check	Verified	Words
C01	direction: number of vessels coloured by fluoroscopy: 0 vessels to 2 vessels (whole model) *shape: steady (equal steps); 100% of patients same; verified; risk saturates near 100%*	18% to 99%	log-odds 5.89, OR 362; range across patients 5.83 to 5.83	242	train; test unclear (AUC gain 0.041 [-0.032, 0.123], n=61)	yes	on the training data only
C02	direction: slope of the ST segment at peak exercise: an upsloping ST segment to a flat ST segment (whole model) *shape: categorical; highest to lowest: a flat ST segment, a downsloping ST segment, an upsloping ST segment*	18% to 91%	log-odds 3.81, OR 44.9; range across patients 3.81 to 3.81	115, 110	train; test disagrees (AUC gain -0.011 [-0.065, 0.036], n=61)	yes	on the training data only
C03	direction: slope of the ST segment at peak exercise: an upsloping ST segment to a flat ST segment (direct contribution only, not through any part) *shape: categorical; highest to lowest: a flat ST segment, a downsloping ST segment, an upsloping ST segment*	18% to 87%	log-odds 3.46, OR 31.9; range across patients 3.46 to 3.46	115, 110	train; test disagrees (AUC gain -0.012 [-0.059, 0.030], n=61)	yes	on the training data only
C04	direction: number of vessels coloured by fluoroscopy: 0 vessels to 2 vessels (direct contribution only, not through any part) *shape: steady (equal steps); 100% of patients same; verified*	18% to 80%	log-odds 2.91, OR 18.3; range across patients 2.91 to 2.91	242	train; test disagrees (AUC gain -0.001 [-0.039, 0.032], n=61)	yes	on the training data only
C05	direction: thallium stress test result: a normal thallium scan to thallium stress test result other than a normal thallium scan, a fixed defect on the thallium scan (whole model) *shape: categorical; highest to lowest: thallium stress test result other than a normal thallium scan, a fixed defect on the thallium scan, a normal thallium scan, a fixed defect on the thallium scan*	18% to 55%	log-odds 1.74, OR 5.71; range across patients 1.74 to 1.74	135, 92	train; test unclear (AUC gain 0.012 [-0.017, 0.044], n=61)	yes	on the training data only
C06	direction: ST depression on exercise: 0.0 mm to 2.8 mm (whole model) *shape: hinge at 1.0 mm; 100% of patients same; fallback; matches the fold at 1.0 mm*	18% to 55%	log-odds 0.87, OR 2.38; range across patients 0.00 to 1.74	242	train; test unclear (AUC gain 0.005 [-0.005, 0.022], n=61)	yes	on the training data only
C07	direction: serum cholesterol: 193 mg/dL (5.0 mmol/L) to 308 mg/dL (8.0 mmol/L) (whole model) *shape: steady; 100% of patients same; verified*	8% to 45%	log-odds 2.22, OR 9.23; range across patients 2.22 to 2.22	242	train; test disagrees (AUC gain -0.022 [-0.066, 0.011], n=61)	yes	on the training data only
C08	direction: resting blood pressure: 110 mm Hg to 155 mm Hg (whole model) *shape: steady; 100% of patients same; verified*	9% to 35%	log-odds 1.65, OR 5.20; range across patients 1.65 to 1.65	242	train; test unclear (AUC gain 0.000 [-0.022, 0.020], n=61)	yes	on the training data only
C09	direction: chest pain type: no chest pain (asymptomatic) to typical angina (whole model) *shape: categorical; highest to lowest: no chest pain (asymptomatic), atypical angina, non-anginal chest pain, typical angina*	18% to <1%	log-odds -4.55, OR 0.01; range across patients -4.78 to -4.08	108, 22	train; test agrees (AUC gain 0.041 [0.003, 0.083], n=61)	yes	clearly
C10	direction: sex: male to female (whole model) *shape: categorical; highest to lowest: male, female*	18% to <1%	log-odds -3.94, OR 0.02; range across patients -3.94 to -3.94	159, 83	train; test unclear (AUC gain 0.023 [-0.014, 0.068], n=61)	yes	on the training data only
C11	direction: chest pain type: no chest pain (asymptomatic) to non-anginal chest pain (whole model) *shape: categorical; highest to lowest: no chest pain (asymptomatic), atypical angina, non-anginal chest pain, typical angina*	18% to 1%	log-odds -3.43, OR 0.03; range across patients -3.43 to -3.43	108, 70	train; test agrees (AUC gain 0.041 [0.003, 0.083], n=61)	yes	clearly
C12	direction: chest pain type: no chest pain (asymptomatic) to chest pain type other than atypical angina, no chest pain (asymptomatic) (direct contribution only, not through any part) *shape: categorical; highest to lowest: no chest pain (asymptomatic), atypical angina, chest pain type*	18% to 2%	log-odds -2.51, OR 0.08; range across patients -2.51 to -2.51	108, 92	train; test agrees (AUC gain 0.037 [0.006, 0.074], n=61)	yes	clearly

Claim	What is checked	Risk, typical patient	Estimate (log-odds; OR); range across patients	n	Test check	Verified	Words
C13	direction: age: 42 years to 66 years (whole model) *shape: steady; 100% of patients same; verified*	10% to 25%	log-odds 1.05, OR 2.87; range across patients 1.05 to 1.05	242	train; test disagrees (AUC gain -0.008 [-0.018, 0.001], n=61)	yes	on the training data only
C14	direction: thallium stress test result: a normal thallium scan to a fixed defect on the thallium scan (whole model) *shape: categorical; highest to lowest: thallium stress test result other than a normal thallium scan, a fixed defect on the thallium scan, a normal thallium scan, a fixed defect on the thallium scan*	18% to 3%	log-odds -1.83, OR 0.16; range across patients -1.83 to -1.83	135, 15 (small n)	train; test unclear (AUC gain 0.012 [-0.017, 0.044], n=61)	yes	the model has little data on this; on the training data only
C15	direction: slope of the ST segment at peak exercise: an upsloping ST segment to a downsloping ST segment (direct contribution only, not through any part) *shape: categorical; highest to lowest: a flat ST segment, a downsloping ST segment, an upsloping ST segment*	18% to 32%	log-odds 0.77, OR 2.16; range across patients 0.77 to 0.77	115, 17 (small n)	train; test disagrees (AUC gain -0.012 [-0.059, 0.030], n=61)	yes	the model has little data on this; on the training data only
C16	direction: slope of the ST segment at peak exercise: an upsloping ST segment to a downsloping ST segment (whole model) *shape: categorical; highest to lowest: a flat ST segment, a downsloping ST segment, an upsloping ST segment*	18% to 32%	log-odds 0.77, OR 2.16; range across patients 0.77 to 0.77	115, 17 (small n)	train; test disagrees (AUC gain -0.011 [-0.065, 0.036], n=61)	yes	the model has little data on this; on the training data only
C17	direction: chest pain type: no chest pain (asymptomatic) to atypical angina (direct contribution only, not through any part) *shape: categorical; highest to lowest: no chest pain (asymptomatic), atypical angina, chest pain type other than atypical angina, no chest pain (asymptomatic)*	18% to 5%	log-odds -1.34, OR 0.26; range across patients -1.34 to -1.34	108, 42	train; test agrees (AUC gain 0.037 [0.006, 0.074], n=61)	yes	clearly
C18	direction: chest pain type: no chest pain (asymptomatic) to atypical angina (whole model) *shape: categorical; highest to lowest: no chest pain (asymptomatic), atypical angina, non-anginal chest pain, typical angina*	18% to 5%	log-odds -1.34, OR 0.26; range across patients -1.34 to -1.34	108, 42	train; test agrees (AUC gain 0.041 [0.003, 0.083], n=61)	yes	clearly
C19	direction: fasting blood sugar: fasting blood sugar of 120 mg/dL or less to fasting blood sugar above 120 mg/dL (whole model) *shape: categorical; highest to lowest: fasting blood sugar of 120 mg/dL or less, fasting blood sugar above 120 mg/dL*	18% to 8%	log-odds -0.92, OR 0.40; range across patients -0.92 to -0.92	209, 33	train; test disagrees (AUC gain -0.001 [-0.010, 0.009], n=61)	yes	on the training data only
C20	direction: maximum heart rate: 118 beats per minute to 175 beats per minute (whole model) *shape: hinge at 145 beats per minute; 100% of patients same; verified; matches the fold at 145 beats per minute*	19% to 14%	log-odds -0.36, OR 0.70; range across patients -0.36 to -0.36	242	train; test unclear (AUC gain 0.002 [0.000, 0.009], n=61)	yes	slightly; on the training data only
C21	direction: maximum heart rate: 118 beats per minute to 175 beats per minute (direct contribution only, not through any part) *shape: flat; 100% of patients same; verified*	18% to 18%	log-odds 0.005, OR 1.00; range across patients 0.005 to 0.005	242	train; test unclear (AUC gain 0.000 [0.000, 0.000], n=61)	yes	slightly; on the training data only
C22	direction: resting ECG: a normal resting ECG to resting ECG other than a normal resting ECG (whole model) *shape: categorical; highest to lowest: a normal resting ECG, resting ECG other than a normal resting ECG*	18% to 18%	log-odds -0.46, OR 0.63; range across patients -1.74 to 0.00	121, 121	train; test disagrees (AUC gain -0.008 [-0.021, 0.000], n=61)	yes	slightly; on the training data only
C23	magnitude: number of vessels coloured by fluoroscopy: 0 vessels to 2 vessels, typical patient *shape: steady (equal steps); 100% of patients same; verified; risk saturates near 100%*	18% to 99%	log-odds 5.83, OR 340	242	train; test unclear (AUC gain 0.041 [-0.032, 0.123], n=61)	yes	on the training data only
C24	magnitude: slope of the ST segment at peak	18% to 91%	log-odds 3.81, OR 44.9	115, 110	train; test disagrees (AUC gain -0.011	yes	on the training

Claim	What is checked	Risk, typical patient	Estimate (log-odds; OR); range across patients	n	Test check	Verified	Words
C25	magnitude: thallium stress test result: a normal thallium scan to thallium stress test result other than a normal thallium scan, a fixed defect on the thallium scan, typical patient *shape: categorical; highest to lowest: thallium stress test result other than a normal thallium scan, a fixed defect on the thallium scan, a normal thallium scan, a fixed defect on the thallium scan*	18% to 55%	log-odds 1.74, OR 5.71	135, 92	train; test unclear (AUC gain 0.012 [-0.017, 0.044], n=61)	yes	on the training data only
C26	magnitude: ST depression on exercise: 0.0 mm to 2.8 mm, typical patient *shape: hinge at 1.0 mm; 100% of patients same; fallback; matches the fold at 1.0 mm*	18% to 55%	log-odds 1.74, OR 5.68	242	train; test unclear (AUC gain 0.005 [-0.005, 0.022], n=61)	yes	on the training data only
C27	magnitude: serum cholesterol: 193 mg/dL (5.0 mmol/L) to 308 mg/dL (8.0 mmol/L), typical patient *shape: steady; 100% of patients same; verified*	8% to 45%	log-odds 2.22, OR 9.23	242	train; test disagrees (AUC gain -0.022 [-0.066, 0.011], n=61)	yes	on the training data only
C28	null: model ignores exercise-induced angina (exact rule)		max change 0.000 (max |change in log-odds|)	242	train	yes	
C29	structure: made of ca_slope2_relay, cp3_relay, st_depression_normal_ecg_gate, max_heart_rate_hinge, typical_angina_ca_switch and inputs age, sex, cp, trestbps, chol, fbs, thalach, slope, ca, thal; added		log-odds n/a, OR n/a	242	train	yes	
C30	direction: number of vessels coloured by fluoroscopy: 0 vessels to 2 vessels (through this part) *shape: steady (equal steps); 100% of patients same; verified*	18% to 61%	log-odds 2.04, OR 7.73; range across patients 1.98 to 1.98	242	train; test disagrees (AUC gain -0.003 [-0.032, 0.019], n=61)	yes	on the training data only
C31	direction: chest pain type: chest pain type other than typical angina to typical angina (through this part) *shape: categorical; highest to lowest: chest pain type other than typical angina, typical angina*	18% to 2%	log-odds -2.04, OR 0.13; range across patients -2.26 to -1.57	220, 22	train; test unclear (AUC gain 0.001 [-0.005, 0.008], n=61)	yes	on the training data only
C32	interaction: effect of number of vessels coloured by fluoroscopy (0 vessels to 2 vessels) by chest pain type: typical angina minus chest pain type other than typical angina; stronger in typical angina *shape: steady (equal steps); 100% of patients same; verified; risk saturates near 0%*	typical angina: <1% to 55%; chest pain type other than typical angina: 18% to 99%	log-odds difference 0.70, OR 2.00	22, 220	train; test agrees (AUC gain 0.001 [-0.005, 0.008], n=61); group difference on test 0.70 [0.70, 0.70]	yes	clearly
C33	structure: its formula reproduces it on 242 training patients (largest difference 0.0)		log-odds 0.00, OR n/a	242	train	yes	
C34	direction: ST depression on exercise: 0.0 mm to 2.8 mm (through this part) *shape: hinge at 1.0 mm; 100% of patients same; fallback; one point with C37 (median fold 1.0 mm, typical patient 1.0 mm, 10th-90th percentile 1.0 mm to 1.0 mm)*	18% to 55%	log-odds 0.87, OR 2.38; range across patients 0.00 to 1.74	242	train; test unclear (AUC gain 0.005 [-0.005, 0.022], n=61)	yes	on the training data only
C35	direction: resting ECG: a normal resting ECG to resting ECG other than a normal resting ECG (through this part) *shape: categorical; highest to lowest: a normal resting ECG, resting ECG other than a normal resting ECG*	99% to 98% (typical patient this part acts on)	log-odds -0.46, OR 0.63; range across patients -1.74 to 0.00	121, 121	train; test disagrees (AUC gain -0.008 [-0.021, 0.000], n=61)	yes	slightly; on the training data only
C36	interaction: effect of ST depression on exercise (0.0 mm to 2.8 mm) by resting ECG: resting ECG other than a normal resting ECG minus a normal resting ECG; stronger in a normal resting ECG *shape: hinge at 1.0 mm; 100% of patients same; fallback*	resting ECG other than a normal resting ECG: 18% to 18%; a normal resting ECG: 18% to 55%	log-odds difference -1.74, OR 0.18	121, 121	train; test agrees (AUC gain -0.008 [-0.021, 0.000], n=61); group difference on test -1.74 [-1.74, -1.74]	yes	clearly

Claim	What is checked	Risk, typical patient	Estimate (log-odds; OR); range across patients	n	Test check	Verified	Words
C37	threshold: ST depression on exercise: fold at about 1.0 mm (starts) for patients with a normal resting ECG *shape: hinge at 1.0 mm; 100% of patients same; fallback; one point with C37 (median fold 1.0 mm, typical patient 1.0 mm, 10th-90th percentile 1.0 mm to 1.0 mm)*	13% to 63% (typical patient with a fold (patients with a normal resting ECG))	fold 1.01 mm (IQR 1.01 to 1.01); past it log-odds 2.41, OR 11.1; range across patients 2.41 to 2.41	121, 41	train; test unclear (AUC gain 0.005 [-0.005, 0.022], n=61)	yes	on the training data only
C38	subgroup: who this part acts on: 42 of 242 patients	98% to 99% (typical patient this part acts on (risk without the part, then with it))	share 0.17; log-odds 1.08, OR 2.95; range across patients 0.19 to 2.50	42, 200	train; test unclear (AUC gain 0.005 [-0.006, 0.025], n=61)	yes	slightly; on the training data only
C39	structure: its formula reproduces it on 242 training patients (largest difference 2.22045e-16)		log-odds 2e-16, OR n/a	242	train	yes	
C40	direction: number of vessels coloured by fluoroscopy: 0 vessels to 2 vessels (through this part) *shape: steady (equal steps); 100% of patients same; verified*	18% to 36%	log-odds 0.94, OR 2.56; range across patients 0.94 to 0.94	242	train; test disagrees (AUC gain -0.002 [-0.019, 0.011], n=61)	yes	on the training data only
C41	direction: slope of the ST segment at peak exercise: slope of the ST segment at peak exercise other than a flat ST segment to a flat ST segment (through this part) *shape: categorical; highest to lowest: a flat ST segment, slope of the ST segment at peak exercise other than a flat ST segment*	18% to 23%	log-odds 0.34, OR 1.41; range across patients 0.34 to 0.34	132, 110	train; test disagrees (AUC gain -0.003 [-0.016, 0.005], n=61)	yes	on the training data only
C42	structure: its formula reproduces it on 242 training patients (largest difference 2.22045e-16)		log-odds 2e-16, OR n/a	242	train	yes	
C43	direction: chest pain type: chest pain type other than non-anginal chest pain to non-anginal chest pain (through this part) *shape: categorical; highest to lowest: chest pain type other than non-anginal chest pain, non-anginal chest pain*	18% to 8%	log-odds -0.92, OR 0.40; range across patients -0.92 to -0.92	172, 70	train; test unclear (AUC gain 0.001 [-0.008, 0.013], n=61)	yes	on the training data only
C44	structure: its formula reproduces it on 242 training patients (largest difference 0.0)		log-odds 0.00, OR n/a	242	train	yes	
C45	direction: maximum heart rate: 118 beats per minute to 175 beats per minute (through this part) *shape: hinge at 145 beats per minute; 100% of patients same; verified; one point with C46 (median fold 145 beats per minute, typical patient 145 beats per minute, 10th-90th percentile 145 beats per minute to 145 beats per minute)*	19% to 14%	log-odds -0.36, OR 0.70; range across patients -0.36 to -0.36	242	train; test unclear (AUC gain 0.002 [0.000, 0.009], n=61)	yes	slightly; on the training data only
C46	threshold: maximum heart rate: fold at about 145 beats per minute (starts) *shape: hinge at 145 beats per minute; 100% of patients same; verified; one point with C46 (median fold 145 beats per minute, typical patient 145 beats per minute, 10th-90th percentile 145 beats per minute to 145 beats per minute)*	19% to 13% (typical patient with a fold)	fold 145 beats per minute (IQR 145 to 145); past it log-odds -0.44, OR 0.64; range across patients -0.44 to -0.44	242, 155	train; test unclear (AUC gain 0.002 [0.000, 0.009], n=61)	yes	on the training data only
C47	structure: its formula reproduces it on 242 training patients (largest difference 0.0)		log-odds 0.00, OR n/a	242	train	yes	

Not supported by the model. Every claim passed its checks.

1.2 Coverage

Coverage is computed by the current code (the function behind the `get_coverage` tool) under these conventions. *Candidate nodes* are all nodes of the explained model state except the output nodes. A candidate node is *covered* when some annotation of the hierarchy (a leaf, or a composition together with its descendants) contains it and all its enabled outgoing edges. *Exact Null* parts, which have no effect on any output for any input, count as covered: nodes with no path to an output over enabled, non-zero-weight edges (disconnected inputs and dead ends), and edges that have zero weight or touch such a node. Compositional coverage is the fraction of required compositions present: every annotation except a single global root containing every output must have a parent; a parentless, childless annotation made only of Null nodes is exempt.

- **Null, disconnected input (9):** `-10` (fbs:0), `-13` (restecg:1), `-14` (restecg:2), `-16` (exang:0), `-17` (exang:1), `-19` (slope:1), `-24` (thal:4), `-26` (thal:7), `-3` (sex:1).
- **Uncovered nodes:** none.
- **Uncovered edges:** none.
- **Missing compositions:** none.

1.3 Annotation hierarchy

The hierarchy has 1 root; the global root is heart_disease_logit. Tags (A1, A2, ...) are used in the figures. ▷ A1: heart_disease_logit (composition, section 1.6.1)

- ▷ A2: ca_slope2_relay (leaf, section 1.6.2)
- ▷ A3: cp3_relay (leaf, section 1.6.3)
- ▷ A4: st_depression_normal_ecg_gate (leaf, section 1.6.4)
- ▷ A5: max_heart_rate_hinge (leaf, section 1.6.5)
- ▷ A6: typical_angina_ca_switch (leaf, section 1.6.6)

1.4 Network figures

1.5 Refactoring operations

The explained model state is the phenotype after these operations, replayed in order. All are function-preserving; the largest change in the model output over the 242 training rows is 3.33×10^{-16} (tolerance 1.00×10^{-9}).

- #26: split `-22` into `-22_a`, `-22_b`, `-22_c`.
- #27: split `-20` into `-20_a`, `-20_b`.
- #28: split `-15` into `-15_a`, `-15_b`.

1.6 Annotations

1.6.1 A1: heart_disease_logit

Kind: Composition annotation (depth 0); children A2 (section 1.6.2), A3 (section 1.6.3), A4 (section 1.6.4), A5 (section 1.6.5), A6 (section 1.6.6).

- **Entry nodes:** `-1` (age), `-2` (sex:0), `-3` (sex:1), `-5` (cp:2), `-7` (cp:4), `-8` (trestbps), `-9` (chol), `-10` (fbs:0), `-11` (fbs:1), `-13` (restecg:1), `-14` (restecg:2), `-15_a` (thalach_a), `-16` (exang:0), `-17` (exang:1), `-19` (slope:1), `-20_a` (slope:2_a), `-21` (slope:3), `-22_a` (ca_a), `-23` (thal:3), `-24` (thal:4), `-25` (thal:6), `-26` (thal:7).
- **Exit nodes:** `0`.

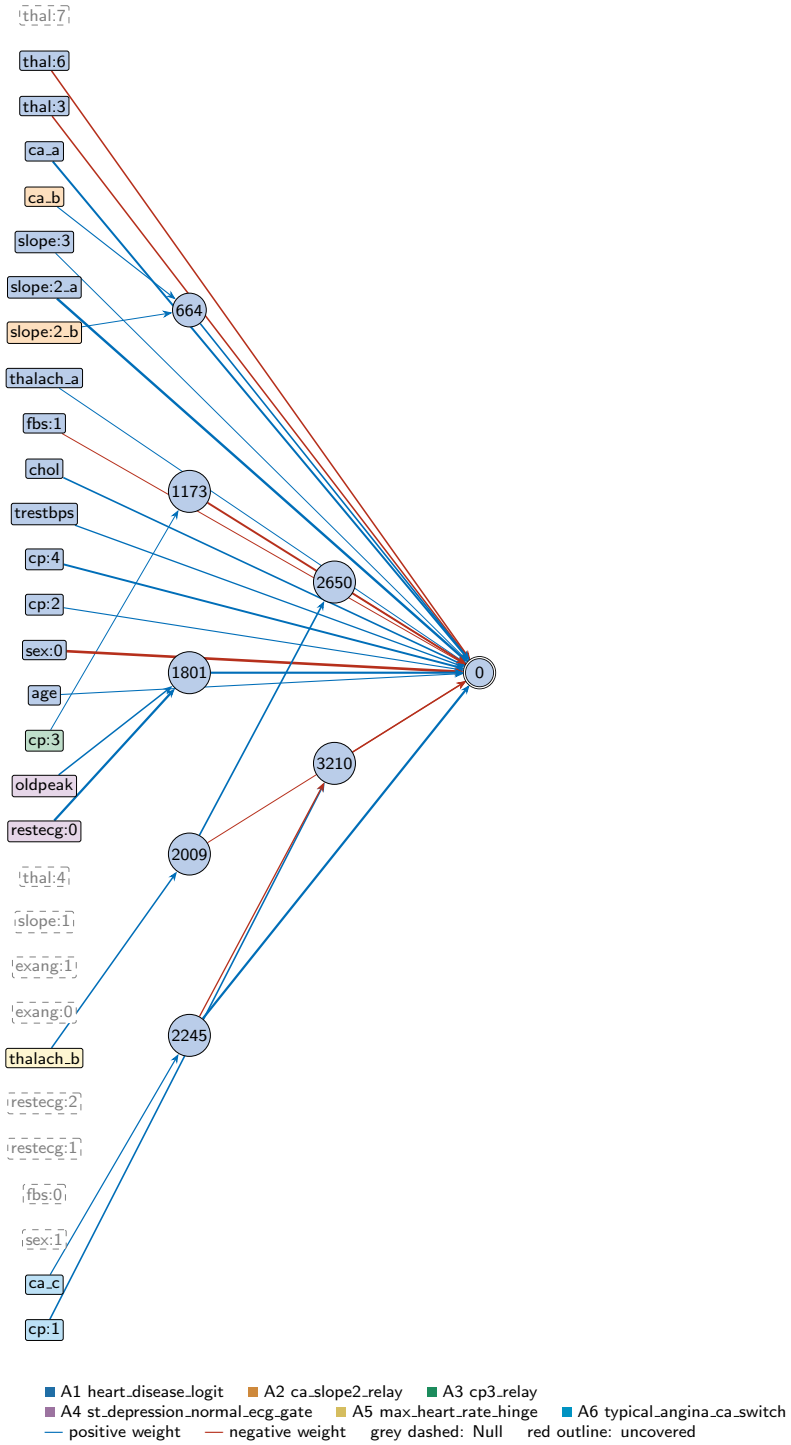


Figure 1: heart-s1-c1: full view of the explained model state.

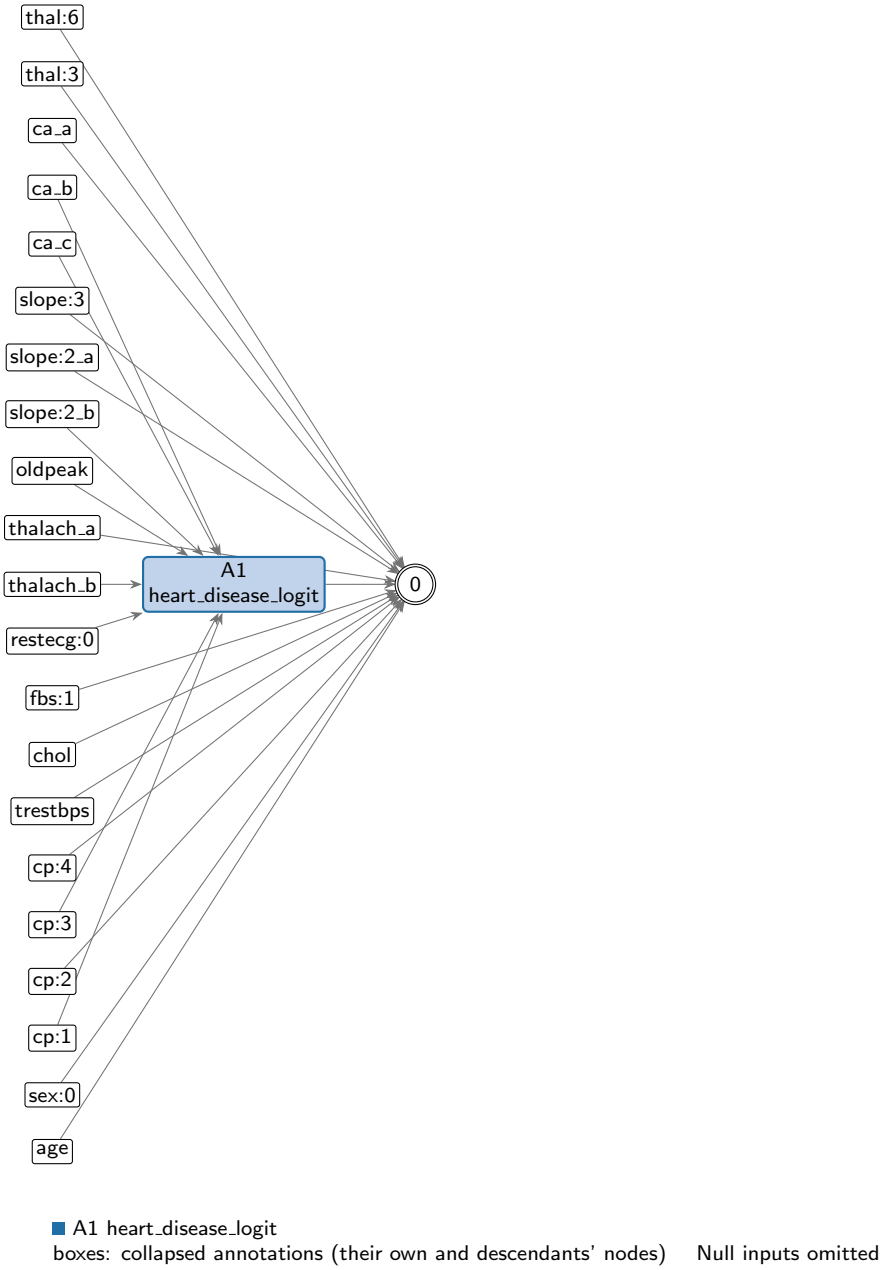


Figure 2: heart-s1-c1: collapsed view at hierarchy level 0 (fully recomposed).

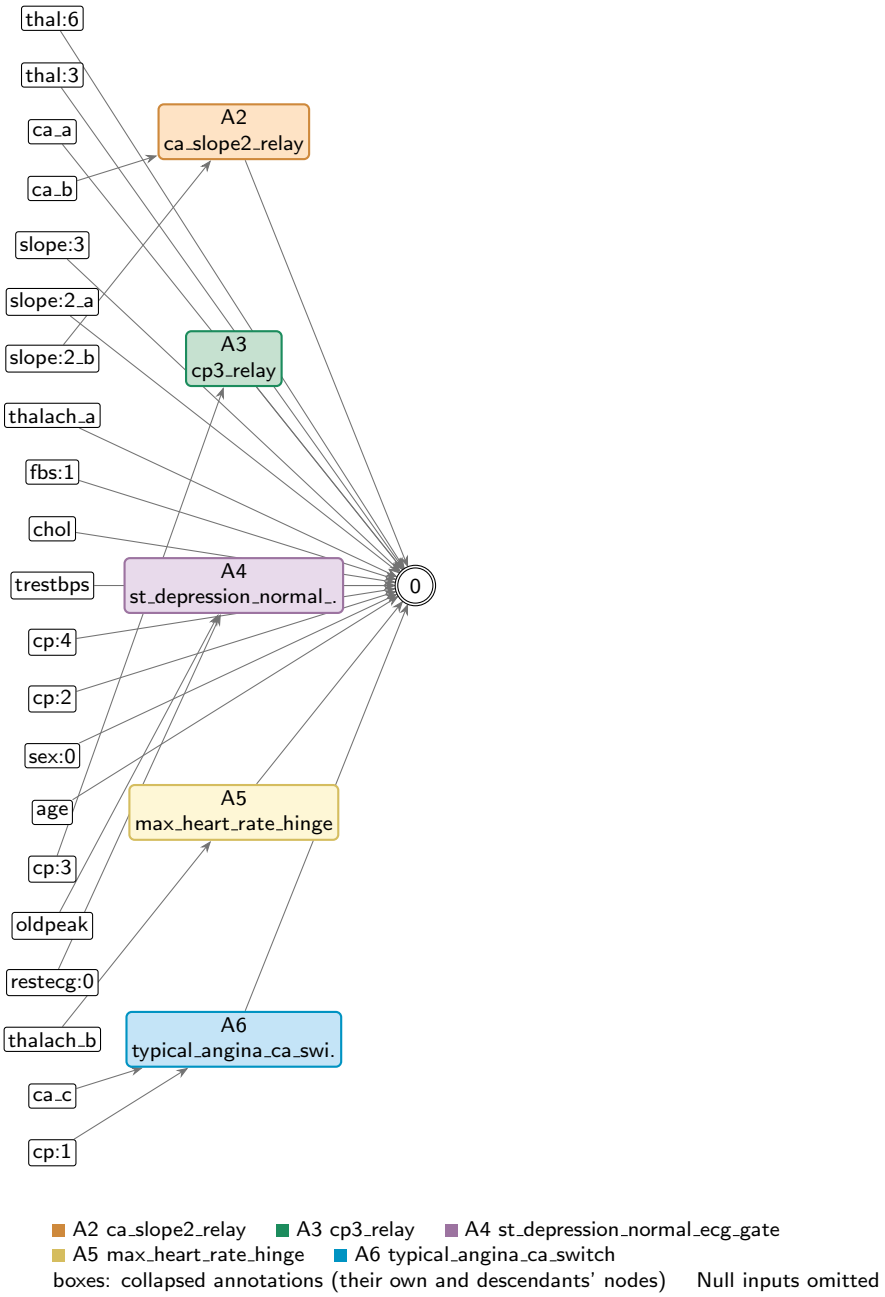


Figure 3: heart-s1-c1: collapsed view at hierarchy level 1.

- **Internal nodes:** none.
- **Edges (weight):** `-1`→`0` (0.399), `-2`→`0` (-1.87), `-5`→`0` (0.443), `-7`→`0` (1.25), `-8`→`0` (0.665), `-9`→`0` (0.931), `-11`→`0` (-0.315), `-21`→`0` (0.197), `-23`→`0` (-0.866), `-25`→`0` (-0.861), `664`→`0` (0.92), `1173`→`0` (-1.24), `1801`→`0` (1.4), `2009`→`0` (-0.151), `2245`→`0` (1.53), `2650`→`0` (-0.172), `3210`→`0` (-0.833), `-22_a`→`0` (1.3), `-20_a`→`0` (1.72), `-15_a`→`0` (0.00188).
- **With its descendants:** 38 nodes.
- **Covers** (coverage, own and as part of the hierarchy): 37 candidate nodes.

Hypothesis. Hypothesis: the heart-disease logit is essentially a logistic regression. 13 direct input->output edges (largest: sex:0 -1.87, slope:2 +1.72, ca +1.30, cp:4 +1.25, chol +0.93, thal:3 -0.87, thal:6 -0.86, trestbps +0.67, cp:2 +0.44, age +0.40, fbs:1 -0.32, slope:3 +0.20, thalach +0.002) are added to five children: ca_slope2_relay and the 2245 relay (raise the effective ca weight to 2.60 per SD), cp3_relay (non-anginal pain lowers risk), st_depression_normal_ecg_gate (ST depression raises risk only with a normal resting ECG), typical_angina_ca_switch's 3210 (typical angina lowers risk, less with more vessels) and max_heart_rate_hinge (small). Nine inputs are unconnected (sex:1, fbs:0, restecg:1, restecg:2, exang:0, exang:1, slope:1, thal:4, thal:7): exercise-induced angina is ignored. Evidence (train, n=242): AUC 0.9392; all hidden units replaced by their best linear fits 0.9355 (= L2 logistic regression 0.9355); all ReLUs as identity 0.9078; all hidden mean-ablated 0.9241. The gain over linear (+0.0037) comes from the ECG/ST gate (+0.0034 restored alone), the heart-rate hinge (+0.0008) and the typical-angina switch (+0.0003); the relays contribute none. Direct edges carry logit SD 4.25 of 5.63.

Closed form (children as named functions):

$$
\begin{aligned}
y_0 &= \sigma\big(z_0\big)\\
z_0 &= -0.276 + 1.3x_{ca_a} + 0.666x_{trestbps}\\
&\quad + 0.92\,\mathrm{ca_slope2_relay}\big(x_{ca_b}, x_{slope:2_b}\big)\\
&\quad + 0.00188x_{thalach_a} + 1.25x_{cp:4}\\
&\quad + 1.4\,\mathrm{st_depression_normal_ecg_gate}\big(x_{restecg:0}, x_{oldpeak}\big)\\
&\quad + 1.53\,\mathrm{typical_angina_ca_switch_0}\big(x_{ca_c}, x_{cp:1}\big)\\
&\quad + 0.197x_{slope:3} + 0.399x_{age} + 1.72x_{slope:2_a} + 0.443x_{cp:2} + 0.931x_{chol}\\
&\quad - 1.87x_{sex:0} - 1.24\,\mathrm{cp3_relay}\big(x_{cp:3}\big) - 0.315x_{fbs:1}\\
&\quad - 0.861x_{thal:6} - 0.151\,\mathrm{max_heart_rate_hinge_0}\big(x_{thalach_b}\big)\\
&\quad - 0.172\,\mathrm{max_heart_rate_hinge_1}\big(x_{thalach_b}\big)\\
&\quad - 0.833\,\mathrm{typical_angina_ca_switch_1}\big(x_{ca_c}, x_{cp:1}\big)\\
&\quad - 0.866x_{thal:3}
\end{aligned}
$$

Expanded over the children:

$$
\begin{aligned}
y_0 &= \sigma\big(z_0\big)\\
z_0 &= -0.276 + 1.3x_{ca_a} + 0.666x_{trestbps} + 0.92\,\mathrm{relu}\big(0.783\\
&\quad + 0.185x_{slope:2_b} + 0.456x_{ca_b}\big) + 0.00188x_{thalach_a} + 1.25x_{cp:4}\\
&\quad + 1.4\,\mathrm{relu}\big(-1.52 + 1.54x_{restecg:0} + 0.818x_{oldpeak}\big)\\
&\quad + 1.53\,\mathrm{relu}\big(1.28 + 0.578x_{ca_c}\big) + 0.197x_{slope:3}\\
&\quad + 0.399x_{age} + 1.72x_{slope:2_a} + 0.443x_{cp:2} + 0.931x_{chol} - 1.87x_{sex:0}\\
&\quad - 1.24\,\mathrm{relu}\big(1.82 + 0.336x_{cp:3}\big) - 0.315x_{fbs:1}\\
&\quad - 0.861x_{thal:6} - 0.151\,\mathrm{relu}\big(0.201 + 0.833x_{thalach_b}\big)\\
&\quad - 0.172\,\mathrm{relu}\big(0.859 + 0.982\,\mathrm{relu}\big(0.201\\
&\quad + 0.833x_{thalach_b}\big)\big) - 0.833\,\mathrm{relu}\big(0.42 + 0.901x_{cp:1}\\
&\quad - 0.644\,\mathrm{relu}\big(1.28 + 0.578x_{ca_c}\big)\big) - 0.866x_{thal:3}
\end{aligned}
$$

where y_0 is node `0`; z_i, where shown, is the pre-activation of a sigmoid output y_i; weights and biases are shown to 3 significant figures. On the training split (n = 242), the displayed closed form differs from the forward pass by at most 0.00427, and the exact function by at most 4.44×10^{-16}.

Evidence (2 items; full payloads in the JSON record):

- ANALYTICAL, `analytical_formula`; data split: n/a (analytical). Root closed form (get_formula force=True) and effective linear weights through the always-on relays
- EMPIRICAL, `ablation_study`; data split: train. Train split, full sample; ablations and linear replacements on in-memory copies (no graph edits)

Authorship: drafted by AI-assisted draft on 2026-11-10; verified by nobody (unverified).

1.6.2 A2: ca_slope2_relay

Kind: Leaf annotation (depth 1); parent A1 (section 1.6.1).

- **Entry nodes:** `-22_b` (ca_b), `-20_b` (slope:2_b).
- **Exit nodes:** `664`.
- **Internal nodes:** none.
- **Edges (weight):** `-22_b`→`664` (0.456), `-20_b`→`664` (0.185).
- **Covers** (coverage, own and as part of the hierarchy): 2 candidate nodes.

Hypothesis. Hypothesis: node 664 = ReLU(0.456 ca + 0.185 slope:2 + 0.783) is an affine relay that strengthens the direct ca and flat-ST-slope (slope:2) effects. Evidence (train, n=242): pre-activation 0.28..2.19, never clips, so the part is exactly affine on the data; it adds +0.42 logit per SD of ca and +0.17 per SD of slope:2 (output weight 0.92). Mean ablation AUC 0.9392 -> 0.9375; linear replacement leaves AUC at 0.9392. No concept beyond reweighting.

Closed form:

$$y_0 = \mathrm{relu}\big(0.783 + 0.185 x_{slope:2_b} + 0.456 x_{ca_b}\big)$$

where y_0 is node `664`; z_i, where shown, is the pre-activation of a sigmoid output y_i; weights and biases are shown to 3 significant figures. On the training split (n = 242), the displayed closed form differs from the forward pass by at most 4.47×10^{-4}, and the exact function by at most 0.

Evidence (2 items; full payloads in the JSON record):

- ANALYTICAL, `analytical_formula`; data split: train (inferred, inferred from payload keys). Closed form from get_formula (force=True)
- EMPIRICAL, `empirical_viz` (heatmap); data split: train. Train split, full sample (242); local forward passes on in-memory copies for ablations

Authorship: drafted by AI-assisted draft on 2026-11-10; verified by nobody (unverified).

1.6.3 A3: cp3_relay

Kind: Leaf annotation (depth 1); parent A1 (section 1.6.1).

- **Entry nodes:** `-6` (cp:3).
- **Exit nodes:** `1173`.
- **Internal nodes:** none.
- **Edges (weight):** `-6`→`1173` (0.336).

- **Covers** (coverage, own and as part of the hierarchy): 1 candidate node.

Hypothesis. Hypothesis: node 1173 = ReLU(0.336 cp:3 + 1.818) is an always-on relay through which non-anginal chest pain (cp:3) lowers predicted risk. Evidence (train, n=242): pre-activation 1.60..2.34, never clips (entry-exit correlation 1.0); with output weight -1.24 it contributes -0.417 logit per SD of cp:3, about -0.92 logit for non-anginal pain vs other types. cp:3 has no direct edge, so this is its only route. Mean ablation AUC 0.9373; linear replacement 0.9392.

Closed form:

$$y_0 = \mathrm{relu}\big(1.82 + 0.336 x_{cp:3}\big)$$

where y_0 is node `1173`; z_i, where shown, is the pre-activation of a sigmoid output y_i; weights and biases are shown to 3 significant figures. On the training split (n = 242), the displayed closed form differs from the forward pass by at most 0.00246, and the exact function by at most 0.

Evidence (2 items; full payloads in the JSON record):

- ANALYTICAL, `analytical_formula`; data split: train (inferred, inferred from payload keys). Closed form from get_formula (force=True)
- EMPIRICAL, `empirical_viz` (line_plot); data split: train. Train split, full sample (242); local forward passes on in-memory copies for ablations

Authorship: drafted by AI-assisted draft on 2026-11-10; verified by nobody (unverified).

1.6.4 A4: st_depression_normal_ecg_gate

Kind: Leaf annotation (depth 1); parent A1 (section 1.6.1).

- **Entry nodes:** `-12` (restecg:0), `-18` (oldpeak).
- **Exit nodes:** `1801`.
- **Internal nodes:** none.
- **Edges (weight):** `-12`→`1801` (1.54), `-18`→`1801` (0.817).
- **Covers** (coverage, own and as part of the hierarchy): 2 candidate nodes.

Hypothesis. Hypothesis: node 1801 = ReLU(1.542 restecg:0 + 0.817 oldpeak - 1.521) is a gate in which ST depression (oldpeak) raises predicted risk only for patients with a normal resting ECG (restecg:0), once oldpeak is above about its mean (threshold -0.03 SD); with an abnormal resting ECG the threshold is 3.75 SD and the gate is effectively closed. Evidence (train, n=242): active for 41 of 121 normal-ECG patients and 1 of 121 others; adds 1.40 x activation = +1.14 logit per SD oldpeak when open. It is the model's main nonlinearity: mean ablation AUC 0.9339, linear replacement 0.9362, and restoring it onto the all-linear model gains +0.0034 AUC. oldpeak has no direct edge, so the model ignores ST depression for patients with an abnormal resting ECG.

Closed form:

$$y_0 = \mathrm{relu}\big(-1.52 + 1.54 x_{restecg:0} + 0.818 x_{oldpeak}\big)$$

where y_0 is node `1801`; z_i, where shown, is the pre-activation of a sigmoid output y_i; weights and biases are shown to 3 significant figures. On the training split (n = 242), the displayed closed form differs from the forward pass by at most 0.00514, and the exact function by at most 0.

Evidence (2 items; full payloads in the JSON record):

- ANALYTICAL, `analytical_formula`; data split: train (inferred, inferred from payload keys). Closed form from get_formula (force=True)
- EMPIRICAL, `empirical_viz` (heatmap); data split: train. Train split, full sample (242); local forward passes on in-memory copies for ablations

Authorship: drafted by AI-assisted draft on 2026-11-10; verified by nobody (unverified).

1.6.5 A5: max_heart_rate_hinge

Kind: Leaf annotation (depth 1); parent A1 (section 1.6.1).

- **Entry nodes:** `-15_b` (thalach_b).
- **Exit nodes:** `2009`, `2650`.
- **Internal nodes:** none.
- **Edges (weight):** `-15_b`→`2009` (0.833), `2009`→`2650` (0.982).
- **Covers** (coverage, own and as part of the hierarchy): 1 candidate node.

Hypothesis. Hypothesis: a small hinge on maximum heart rate (thalach). Node 2009 = ReLU(0.833 thalach + 0.201) feeds the output (-0.151) and the always-on relay 2650 = ReLU(0.982*2009 + 0.859) (-0.172), a net -0.32 x 2009: higher maximum heart rate lowers predicted risk, but only above about -0.24 SD; below that it has no effect (the direct thalach edge, +0.002, is negligible). Evidence (train, n=242): 2009 active for 64% of patients; logit contribution SD 0.16 (small); mean ablation AUC 0.9391, linear replacement 0.9392, restore-onto-linear gain +0.0008.

Closed form:

$$y_0 = \mathrm{relu}\big(0.201 + 0.833 x_{thalach_b}\big)$$
$$y_1 = \mathrm{relu}\big(0.859 + 0.982\,\mathrm{relu}\big(0.201 + 0.833 x_{thalach_b}\big)\big)$$

where y_0 is node `2009`; y_1 is node `2650`; z_i, where shown, is the pre-activation of a sigmoid output y_i; weights and biases are shown to 3 significant figures. On the training split (n = 242), the displayed closed form differs from the forward pass by at most 7.41×10^{-4}, and the exact function by at most 0.

Evidence (2 items; full payloads in the JSON record):

- ANALYTICAL, `analytical_formula`; data split: train (inferred, inferred from payload keys). Closed form from get_formula (force=True)
- EMPIRICAL, `empirical_viz` (line_plot); data split: train. Train split, full sample (242); local forward passes on in-memory copies for ablations

Authorship: drafted by AI-assisted draft on 2026-11-10; verified by nobody (unverified).

1.6.6 A6: typical_angina_ca_switch

Kind: Leaf annotation (depth 1); parent A1 (section 1.6.1).

- **Entry nodes:** `-22_c` (ca_c), `-4` (cp:1).
- **Exit nodes:** `2245`, `3210`.
- **Internal nodes:** none.
- **Edges (weight):** `-22_c`→`2245` (0.578), `-4`→`3210` (0.901), `2245`→`3210` (-0.644).
- **Covers** (coverage, own and as part of the hierarchy): 2 candidate nodes.

Hypothesis. Hypothesis: (i) node 2245 = ReLU(0.578 ca + 1.276) is an always-on ca relay adding +0.88 logit per SD of ca (output weight 1.528); (ii) node 3210 = ReLU(0.901 cp:1 - 0.644*2245 + 0.420) is a typical-angina switch: it is on only for cp:1 patients and then lowers the logit by 0.833 x its activation, from about -2.26 at the lowest ca (-0.73 SD) to about -1.22 at the highest ca (2.64 SD). So typical angina lowers predicted risk, and the reduction shrinks with more

narrowed vessels. Evidence (train, n=242): 2245 never clips (0.86..2.80); 3210 active for all 22 cp:1 patients and none of the 220 others. Mean ablation AUC 2245 0.9339, 3210 0.9354; linear replacement 0.9392 / 0.9391 (3210 is a cp:1 switch whose within-group response is linear in ca).

Closed form:

$$\begin{aligned} y_0 &= \mathrm{relu}\big(1.28 + 0.578 x_{ca_c}\big) \\ y_1 &= \mathrm{relu}\big(0.42 + 0.901 x_{cp:1} - 0.644\,\mathrm{relu}\big(1.28 \\ &\qquad + 0.578 x_{ca_c}\big)\big) \end{aligned}$$

where y_0 is node `2245`; y_1 is node `3210`; z_i, where shown, is the pre-activation of a sigmoid output y_i; weights and biases are shown to 3 significant figures. On the training split (n = 242), the displayed closed form differs from the forward pass by at most 0.00423, and the exact function by at most 0.

Evidence (2 items; full payloads in the JSON record):

- ANALYTICAL, `analytical_formula`; data split: train (inferred, inferred from payload keys). Closed form from get_formula (force=True)
- EMPIRICAL, `empirical_viz` (heatmap); data split: train. Train split, full sample (242); local forward passes on in-memory copies for ablations

Authorship: drafted by AI-assisted draft on 2026-11-10; verified by nobody (unverified).

1.7 Audit against the definitions

Automatic findings on the stored explanation (codes as in the JSON record's `issues`); they flag what an examiner should check, not verdicts.

- WARNING `unverified`: No human verification is recorded for this explanation.

end of thesis draft